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Abstract

In 2024, Experian reported 450,000 instances of credit card fraud, and this theft is harmful to both individuals who are saving up for life events such as family planning and buying property, and to financial institutions that have to cover the costs. As bad actors leverage technology to be innovative in their scamming, we leveraged two complex deep learning technologies: LSTMs and Transformers. Through research, we recognized that researchers utilize multiple models to strengthen their testing methodology. From Kaggle, we obtained a dataset of over a million rows of synthetic data reflective of bank users and a list of their transactions. Our dataset was reflective of a problem we continued to help mitigate in our model development: class imbalance. Our dataset had less than 1% of fraudulent data. We feature engineered factors such as age, where we flagged that individuals older than 60 were more likely to be victims of fraud schemes. We had one user's transaction history, and by leveraging their historical data, we would then predict if the next transaction on their account was made by the account user or by a third party. We compared the performance of the LSTM and Transformer separately and then created a stacked model to identify if stacking these powerful models provided better results: correct predictions of whether a transaction was real or fraudulent. Overall, though the stacked model wasn't the strongest in all metrics, it produced a strong F1-score: 73%. An F1-score of 73% meant that of all the predictions that this model was making, 73% of them were correct; the stacked model F1-score was significantly better than the F1-score of our LSTM model: 58%. Since we measured using the F1-score due to our imbalanced dataset, we conclude that when the strength of LSTMs in working well with sequential data is combined

with the attention mechanisms of the Transformer, we can obtain stronger results and obtain more accurate detectors of fraud.

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Introduction

The growth of technology has resulted in bad actors being creative in their efforts to scam people. As technology develops at a rapid rate, there remains a discrepancy in financial technology education; people are not aware of all the methods that people employ, by leveraging technology, to scam others. As financial institutions continue to send emails to users pushing for caution when it comes to phone calls, emails, and requests from third parties who may be attempting to scam them, there exist only a limited number of ways to navigate how to identify when scams have occurred. If organizations can navigate ways to mitigate the sheer number of scams from occurring, it would slow the rise of financial fraud. These financial scams can rob people of their savings and their incomes under pretenses by leveraging dishonesty and fear. In this paper, we leverage a dataset of thousands of synthetic data points, consider transaction history to identify if the next transaction is a real transaction that the user performs, or if it is a fraudulent transaction made by an ill-intentioned person. We employed a deep learning methodology, harnessing the strengths of both LSTMs and the famous Transformer from the paper “Attention Is All You Need” that showcased Transformers to the world. Through this paper, we leveraged multiple deep learning models to maximize our chances of fraud detection, and harnessed our knowledge of features to add more features, and compared the results of the LSTM, the Transformer, and the stacked model.

Literature Review

Although it is common nowadays for machine learning algorithms to be applied to this problem, there are many different approaches to detecting credit card fraud that do not involve machine learning. Researchers have shown that statistical methods can also be used to identify credit card fraud. They utilized statistical properties of transaction data to determine suspicious trends [1]. One such example is hypothesis testing, which is a statistical approach that compares genuine and fraudulent transactions using null and alternative hypotheses and statistical tests like t-tests or chi-square tests [2].

However, the significance of machine learning algorithms can not be ignored. There have been countless papers that utilized machine learning in an attempt to solve real-world problems. This approach is still generally favored among researchers for credit card fraud detection today simply due to their ability to adapt and notice complex patterns and related attributes within the dataset [2]. However, with the data collected from credit card transactions, normal and fraudulent transactions are naturally imbalanced, which is often known as class imbalance.

Researchers have generally resolved the issue of class imbalance by performing pre-processing on the dataset, where they would oftentimes oversample fraudulent transactions and under-sample legitimate transactions to balance these two classes [3]. Besides pre-processing, researchers have shown to use the synthetic minority oversampling technique (SMOTE) to resolve imbalanced datasets [3]. Using SMOTE has shown minor improvements to the model's performance. In an experiment that was done in 2018, SMOTE helped researchers achieve a better accuracy by 2-4 percent when compared to other methods of classification [4]. In a different research [5], SMOTE was able to improve the average F1-Score and G-Score. SMOTE has also been seen used within a hybrid resampling technique. Based on this paper [6], the researcher used SMOTE with a modified nearest neighbour method (SMOTE-ENN) to obtain a balanced dataset, which has also been shown to improve the performance of models when compared to not using SMOTE-ENN.

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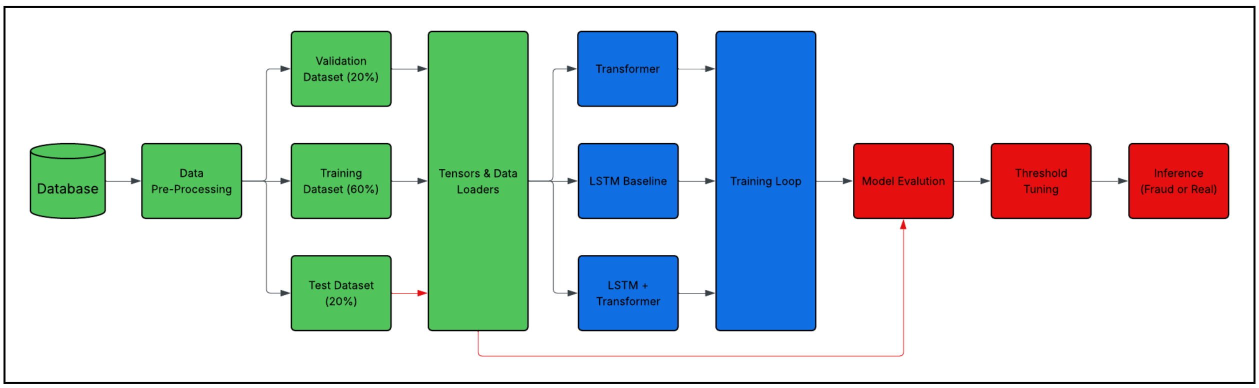
Aside from the various approaches in balancing a dataset, various machine learning algorithms were also used in an attempt to better detect fraudulent transactions. Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), Generative Adversarial Networks (GAN), and Recurrent Neural Networks (RNNs) are just some examples of algorithms that have been used by researchers for credit card fraud detection [2]. For CNNs, they are useful at "classifying images and extracting features from temporal data," which can definitely work well at detecting fraud within transaction sequences [2]. In a separate paper, researchers even proposed a model that uses two algorithms together (LSTM-LMM) to catch fraudulent transactions [7]. In reviewing these papers, models that utilize LSTM tend to appear more often due to their design. Its design enables "long-term dependency on sequence predictions," which is very useful in helping the model remember long-term patterns [7].

Data & Pre-Processing

The dataset that we are using for this paper was obtained from Kaggle [8]. As stated within the description of the dataset, these data were generated using a Sparkov Data Generation rather than actual transactions from bank institutions. Obtaining real-time datasets would be challenging as banks and financial institutions do not expose their customer's data due to the General Data Protection Regulation (GDPR) [3]. GDPR enforces limitations on data collection, processing, and exposure to protect the privacy and security of consumers. Thus, it is only possible to rely on datasets that are generated by simulation as the next possible choice to train, validate, and test the model we are proposing here.

After choosing on a dataset, the dataset was then pre-processed. For the pre-processing, certain pre-existing features were used to generate better features of the data for the model to use. For example, trans_date_trans_time was to generate new labeling such as hour, day of the week, and month of the transaction so that these features can be better processed by the model. We were also able to get the age of the consumer from its date of birth. After obtaining the necessary information in these new dataset that we have changed, we then chose to scale the dataset using the StandardScaler function to standardize the features.

Model Architecture



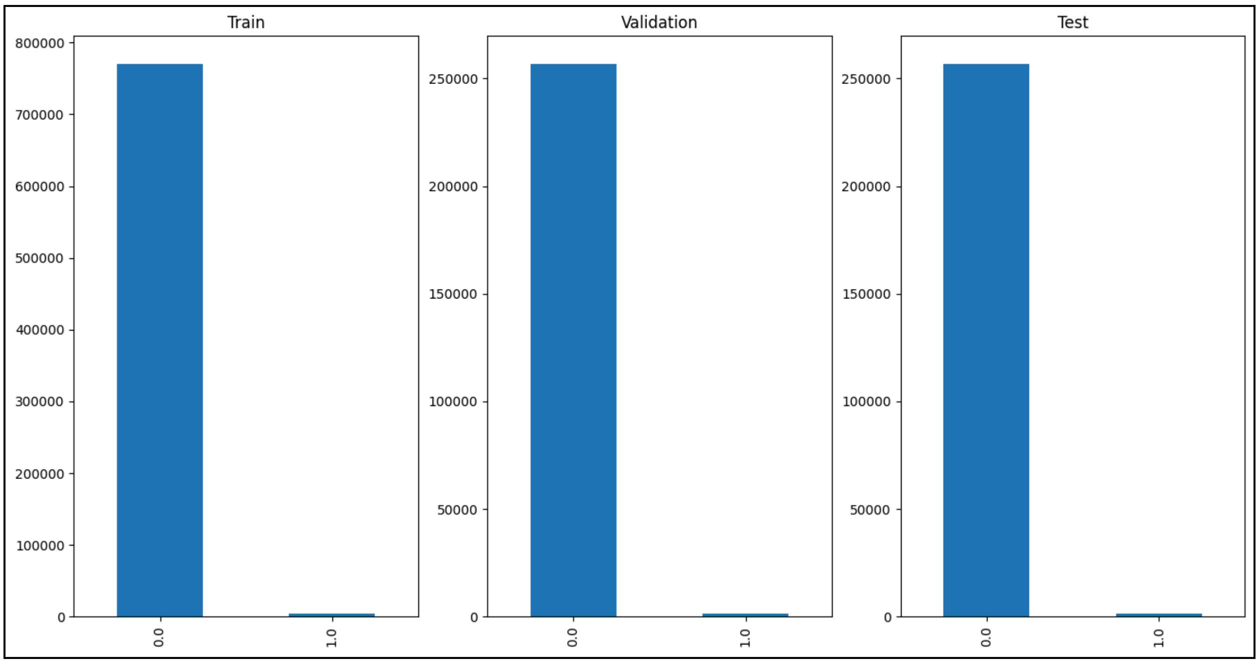
Our proposed workflow in detecting fraudulent transactions can be categorized in three different groups: data pipeline (green), model & training (blue), and evaluation & output (red). After we have pre-processed the dataset, we went for a 60/20/20 split to ensure that the model has enough data to train, validate, and to test the performance of the model. These data are then converted to tensors so that it can be loaded into the models that we are using.

For our models, we went for an LSTM as a baseline to compare against transformers and a model that utilized both of them. To set up the training loop, an Adam optimizer and Binary Cross-Entropy were used. Adam optimizer was used here to update the model's weights during training. On the other hand, Binary Cross-Entropy is used for binary classification, utilizing pos_weight to handle class imbalance. Afterwards, the training loop would run up to 25 epochs.

However, we have also implemented early stopping to detect and stop training once the performance of a model is not improving from training anymore so that time and resources would not be wasted on insignificant improvements.

Model Training

As we identified our models, we recognized that we had a significantly imbalanced dataset, a prevalent problem in this scope of work.



As a result, we chose to optimize using Adam; however, Adam's slow learning rate and ability to update the learning rate individually provides a level of meticulousness that is necessary to preserve the integrity of our largely imbalanced dataset. To navigate a classification problem, we used Binary Cross-Entropy to measure the difference between the model's predicted and actual values. From an overarching perspective, if a fraudulent transaction is not flagged, it could result in significant financial loss; therefore, we used pos_weight to punish the models if they incorrectly determined that a transaction was not fraudulent when it was. Simply, we are aiming to reduce false negatives because in this scenario, false negatives would result in financial loss to the user, and the financial institutions. We leveraged a Scheduler in our models to further control the learning rate; in only our training dataset, having a large learning rate would have led to a lot of false negatives as there are only a few thousand examples of real fraudulent transactions compared to 750,000+ examples of real transactions. To prevent overfitting in our dataset, we leveraged the patience parameter to stop earlier on if the loss began to slow down. Furthermore, as we ran more tests of each model with different numbers of epochs, we recognized that the models would always end at an earlier number of epochs, a sign of convergence. For our training, validation, and testing split, we did not want to have a smaller validation and testing set as they would have fewer, or close to none, examples of fraud behavior. As we trained our model, we prioritized tuning to reduce the impact of the class imbalance on our model. Our goal was to leverage optimizers and schedulers to control the learning weight to ensure that, despite the few fraudulent transactions, our model was still able to learn from them.

Evaluation + Results

To evaluate our models, we did not leverage accuracy; accuracy was very high for all of our models. However, we emphasize that a metric like accuracy is not complex enough to grasp the details of how well or weakly these models perform. Though all three models were able to have a strategy in their predictions instead of random guessing, as seen in Table 1 and the ROC-AUC curve graph, we are not impressed by the performance here, as it does not reflect any valuable information about the imbalance of the classes. Since our classes are so imbalanced and less than 1% of transactions are fake, we used F1-score as our primary unit of measure. The LSTM was the second-best amongst our models; it was close to our stacked model, but would have a low precision because it was generating a lot of false positives. Our goal instead was to reduce the number of false negatives; when transactions are incorrectly labeled as real, the user can lose thousands or millions of dollars in an instant, and we aim to reduce this significantly. Our fusion model was better at precision and did not raise false positives compared to the LSTM itself. Overall, since we reference the F1 - score, we can see in Table 1 that the stacked model's F1-score is the highest, showcasing that 73% of its predictions are correct.

Table 1:

Model	**ROC-AUC**	**AVG Precision**	**F1**	**RECALL**	**PRECISION**
LSTM	**0.953**	**0.746**	**0.58**	**0.83**	**0.45**
TSF	**0.943**	**0.677**	**0.82**	**0.82**	**0.42**
LSTM TSF	**0.945**	**0.749**	**0.73**	**0.78**	**0.68**

ROC - AUC CURVE

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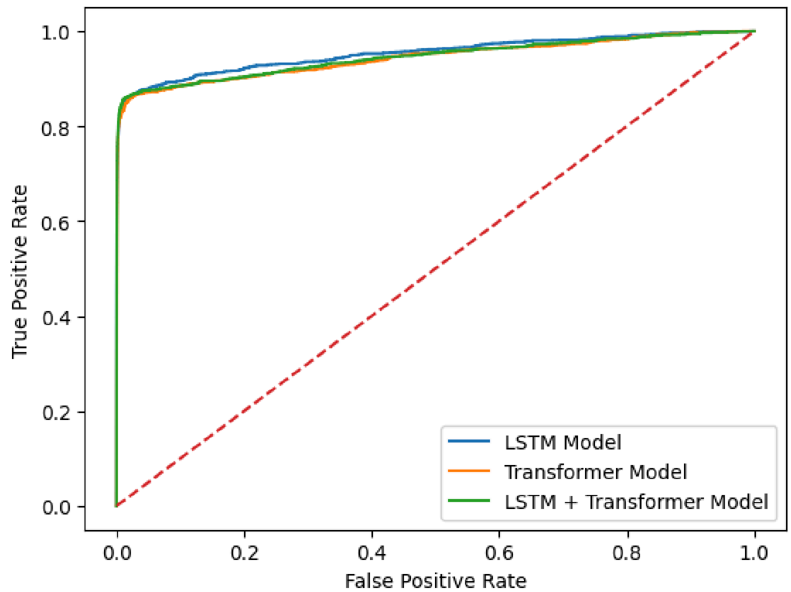

Discussion

Although we have shown that our model can detect fraudulent transactions at a high level of accuracy, more can definitely be done to improve our model's performance. When comparing our results to the results done by [7], we were not even close as their LSTM-RNN model with attention mechanism was able to achieve 1 across precision, recall, F-measure, and accuracy metrics. It really goes to how using other forms of machine learning algorithms together can improve the model's overall performance. Besides model adjustments, techniques like SMOTE to help with class imbalance would help as well.

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